SECTION 4 · SUBSECTION 4.2
STATE, ERRORS AND RETRIES

4.2.1 INTRODUCTION.

Failure handling is not uniform. Some faults abort the whole job, some degrade one file, and some are retried inside a client. The dividing line is whether the remaining stages can still produce a truthful document. A missing repository tree cannot be worked around; a single unparsable file can be reported as such; a failed model call can leave one file unexplained. The operator must know which class a given error belongs to before interpreting an incomplete result (figure 4.2-1).

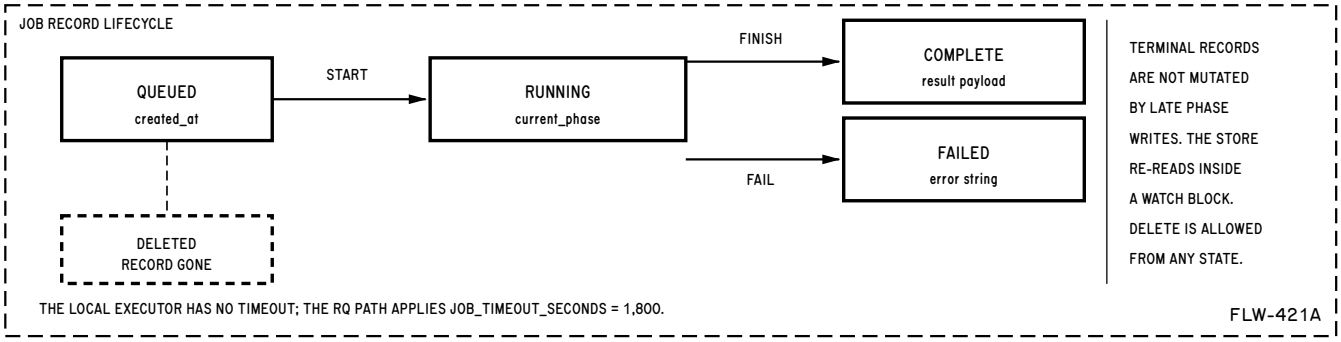


Figure 4.2-1. Queued, Running and Terminal Job States

4.2.2 FUNCTIONAL DESCRIPTION.

In API mode, *pipeline_runner* wraps the entire sequence in one exception handler. Any uncaught exception, including one raised while waiting on a future, is formatted as *Type: message*, the traceback is printed to the server log, and the record moves to FAILED with that string. The HTTP result endpoint then answers 500 for that job. The CLI has no equivalent wrapper: an uncaught exception ends the process, and whatever files were written before the failure remain.

Within the sequence, containment varies. Parse errors are stored in the symbol table with a detail string and later force a CRITICAL risk level. A failed PageRank converges to normalised in-degree, and a total failure yields zero scores rather than an exception. Explainability returns early when the collection or API key is absent and skips individual files whose call returns nothing. The document generator wraps each section, and each output, in its own handler so one bad section does not remove the rest.

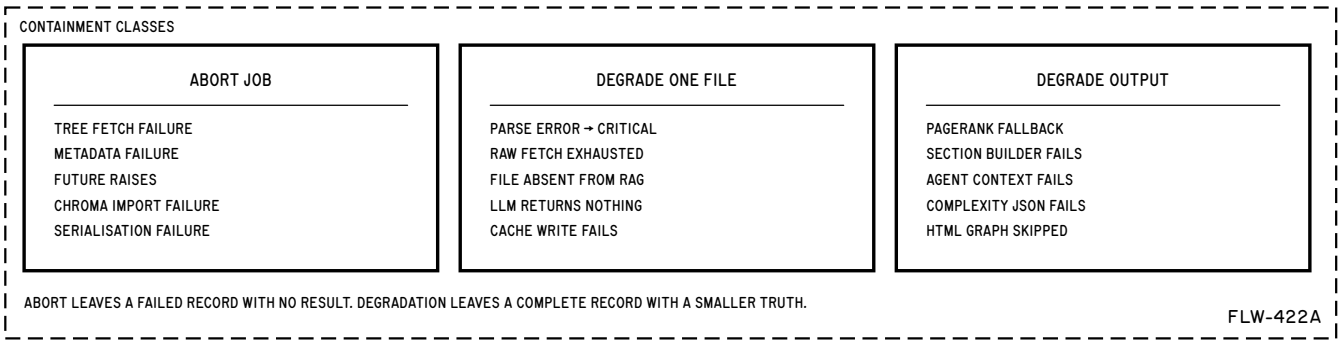


Figure 4.2-2. Which Faults Stop a Run and Which Only Weaken It

4.2.3 PERFORMANCE AND DESIGN DATA.

Table 4.2-1. Retry and Timeout Boundaries

Operation	Attempts	Backoff / limit
Raw file fetch	5	fixed 0.05 s inter-request
GitHub metadata/tree	1	raises to caller
Embedding request	50 texts per call	0.5 s between batches
Chat completion	2	120 s per attempt; 2 s pause
Store WATCH write	5	abort with error
RQ execution	1	1,800 s job timeout
Local thread pool	1	no timeout
Frontend poll	indefinite	1,500 ms; ignores errors

WARNING

Two latent faults escape their intended containment. Incremental chunk deletion calls an undefined *logger*, so a failed delete raises *NameError* instead of being logged. The CLI never writes *agent_context_md* to disk, so a successful run still lacks that deliverable. Neither is exercised by the current suite.

FAILURE CHECK	INVESTIGATOR
1 READ - status and error string	
2 CLASSIFY - abort, file or output	
√ COUNT - explained vs selected	
3 CHECK - parse-error list	
4 CONFIRM - no silent sample	